Quantum Oracle Forensics: strategy and evidence

Team of 3 (beginners in quantum computing). Code: `solver.py`. *All numbers below are copied from* `logs/` *by* `make_report_tables.py` *(fixed seeds, versions in* `logs/versions.json`*). AI-use disclosure:* `AI_DISCLOSURE.md` *(separate page).*

1. Approach

`m` is given, so the problem splits: **m = 1** {constant, balanced, linear, grover, randfun}, **m = 8** {simon, simon2, perm}. The solver keeps an exact **Bayesian posterior** over every (family, secret) hypothesis (m=1: 2 + 510 + 256 + 1 + 7 randfun-grid points; m=8: 1 + 255 + 10,795 subspaces; equal prior per family). Each observation updates it with an *exact* likelihood including both noise types (each table checked against 20,000 simulated shots, `test_solver_model.py`, max error 0.013).

action	what it measures	cost
`bv` (Bernstein-Vazirani/DJ probe)	Walsh index: one repeated string = linear; mostly 0 = constant/grover; P(0)=0.16 randfun, 0 balanced	1
`read(x)`	f(x) classically (needed for b, c, confirmations)	1
`grover(k)`	marked item t	k
`g1`: one Grover step then read f(z)	after one step P(f(z)=1) = $\sin^2(3\theta)$ = $p(3\text{-}4p)^2$: **0.97 for randfun (p=0.3) vs 0.5 for balanced**, 2.2-2.7x the information of two plain reads	2
`probe` (Simon, m=8)	$y \perp s$ (simon), $y \perp H$ (simon2), uniform (perm)	1
`pair(d)`	f(x) vs f(x⊕d): Hamming distance ≤ 1 means d is a period	2

Noise. The solver is not told the noise level, so it is a hidden variable: likelihoods are kept under 4%/2% and 8%/4% with a 50/50 prior and decisions use the posterior marginalised over it. In a paired test on identical oracles this kept accuracy equal (10 oracles won, 10 lost) and saved 1.1 queries on average (95% CI -1.4 to -0.8) (`logs_tau/pairwise_tau.md`).

Choosing the next query. Maximise expected decision-entropy reduction per query cost over {`bv`, `read` at a fresh x, `read` at the best Grover candidate, `g1`, `grover(k)`, k=2..12; m=8: `probe`, `pair` on the two likeliest periods}. Fixed opening: 2 `bv` (m=1) or 3 probes (m=8). *Constant 1 is decided first*: if constant/grover mass ≥ 0.5, up to 6 reads at distinct random x until P(constant 1) leaves (0.005, tau), so it never reaches the Grover branch.

Stopping and confirmations. Stop when the best (family, secret) has posterior ≥ tau and its confirmations are done; otherwise spend the budget and answer the MAP group (constant 0 is therefore never "proved", only made likely).

answer	confirmation (classical reads)
linear (a,b), constant	≥ 2 reads, b = f(x) xor a·x
grover t	2 reads at t returning 1; near-miss candidates (Hamming 1) are counted by the posterior
simon s / simon2 H	2 checks f(x)=f(x⊕s), Hamming ≤ 1 / 1 check per element of H
balanced, randfun, perm	posterior only

2. Statistics and thresholds behind each decision

- **randfun variation**: ones-fraction p ~ $N(0.3, 0.029^2)$ (Binomial(256,0.3)/256), 7-point grid, so one randfun instance cannot be mistaken for a different p by chance.
- **balanced vs randfun** is one sequential test combining the `bv` zero-string rate (0.04-0.08 vs 0.19-0.23), plain reads (0.46-0.48 vs 0.29-0.30 ones) and `g1` bits (0.46-0.48 vs 0.82-0.89).
- **tau = 0.98** by a *pre-registered* paired sweep (40 oracles/family/noise, same oracles for every setting): the lowest tau within 0.5 points of the best accuracy.

tau (mixture, pooled both noises, 640 oracles)	accuracy	mean queries	worst family
0.98	0.9766	30.6	0.900
0.99	0.9797	33.6	0.925
0.995	0.9812	35.5	0.925

Tau sensitivity in the final run (cap 60, same oracles; run *after* the tau choice above and not used to re-tune it):

tau	acc normal	worst fam	saving	acc stress	worst fam	saving
0.95	0.984	0.95	0.592	0.945	0.87	0.535
0.98	0.990	0.98	0.526	0.964	0.91	0.457
0.99	0.994	0.99	0.477	0.969	0.91	0.410

- **perm is the expensive family** because its data (uniform) cannot reveal the noise level and ~11,000 alternatives are only softly penalised: a wrong simon `s` is contradicted by a sample with y·s=1, but readout noise produces such samples from a true oracle with probability 0.04/0.08/0.14/0.24 for |s|=1,2,4,8 (stress noise, readout flip 4%) (likelihood ratio 0.08-0.49 per violating sample, not 0).

- **Grover limit under noise** (exact enumeration of misfire patterns): success per shot after readout, normal/stress: k=4 0.22/0.16, k=6 0.38/0.28, k=8 0.53/0.37, k=12 0.68/0.47 (noise-free k=12 gives 1.0). A k=12 circuit has zero misfires only 61%/37% of the time, so the solver uses k=6-8; a 1-in-256 item is typically found in 2-4 shots, but absence can never be certain.

3. Results (100 oracles per family per noise level)

family	acc normal	mean q normal	acc stress	mean q stress
constant	1.00	33.4	1.00	33.6
balanced	0.98	25.7	0.98	29.7
linear	0.99	5.0	0.99	5.9
grover	0.99	23.0	0.91	29.0
randfun	0.98	26.3	0.91	35.4
simon	1.00	40.0	0.94	46.6
simon2	0.98	24.6	0.98	32.4
perm	1.00	49.8	1.00	51.3
all	**0.990**	**28.5**	**0.964**	**33.0**

Worst-family accuracy: normal 0.98, stress 0.91. Query saving (correct tasks, vs 60): normal 0.526, stress 0.457. Max time per oracle 24.1 s; max qubits 16.

Budget split (mean queries per oracle spent on each action, normal noise; the rest of the 60 is unspent): **constant** bv 2.5, grover 15.7, read 4.5; **balanced** bv 3.3, g1 22.0; **linear** bv 3.4, read 2.3; **grover** bv 2.5, grover 13.3, read 5.3; **randfun** bv 3.3, g1 20.1, grover 0.7, read 0.5; **simon** probe 20.4, read8 18.0; **simon2** probe 18.3, read8 7.3; **perm** probe 26.4, read8 25.1.

Accuracy as a function of the query cap (solver told budget = cap; saving always vs 60):

cap	acc normal	worst fam	mean q	acc stress	worst fam	mean q
20	0.896	0.73	16.5	0.754	0.44	16.8
30	0.958	0.89	21.4	0.875	0.68	23.0
40	0.978	0.94	24.7	0.931	0.86	27.7
50	0.985	0.96	27.1	0.946	0.88	31.0
60	0.990	0.98	28.5	0.964	0.91	33.0

method	acc normal	worst fam	saving	acc stress	worst fam	saving
ours	0.990	0.98	0.526	0.964	0.91	0.457
(1) classical-only (no superposition)	0.552	0.00	0.445	0.481	0.00	0.414
(2) single algorithm (BV/Simon probe only)	0.731	0.00	0.339	0.724	0.00	0.260
(3) random guessing	0.064	0.00	1.000	0.061	0.00	1.000
(4) ours, adaptivity off (fixed schedule, no early stop)	0.944	0.82	0.000	0.886	0.59	0.000

4. Baselines and where classical is enough

Per-family accuracy at cap 60 (normal / stress):

method	constant	balanced	linear	grover	randfun	simon	simon2	perm
full	1.00 / 1.00	0.98 / 0.98	0.99 / 0.99	0.99 / 0.91	0.98 / 0.91	1.00 / 0.94	0.98 / 0.98	1.00 / 1.00
classical	1.00 / 0.99	0.97 / 0.96	1.00 / 1.00	0.23 / 0.20	0.00 / 0.00	0.50 / 0.24	0.54 / 0.41	0.18 / 0.05
single	0.52 / 0.52	1.00 / 0.94	0.50 / 0.50	0.00 / 0.00	0.83 / 0.92	1.00 / 0.92	1.00 / 1.00	1.00 / 0.99
nonadaptive	0.98 / 0.92	0.90 / 0.89	1.00 / 0.99	0.82 / 0.59	0.85 / 0.77	1.00 / 0.94	1.00 / 1.00	1.00 / 0.99

A classical-only strategy (same inference, reads only; birthday search for m=8) reaches 0.55 (normal): it is adequate for constant (1.00), balanced (0.97) and linear (1.00) but fails where structure is hidden in a single input or an XOR period: grover 0.23, simon 0.50, simon2 0.54, perm 0.18. A single quantum circuit reused for everything reaches 0.73: it cannot learn the phase-invisible secrets (constant c, linear b) nor find a grover item (grover 0.00). Switching adaptivity off lowers accuracy by 0.046 (normal) and removes all query saving.

5. Failure analysis

- **normal**: 8 wrong of 800: balanced → randfun ×2; randfun → grover ×2; simon2 → simon ×2; linear (right family, wrong secret) ×1; grover → constant ×1.
- **stress**: 29 wrong of 800: grover → constant ×8; randfun → balanced ×7; simon → perm ×6; balanced → randfun ×2; randfun → grover ×2; simon2 → simon ×2; linear (right family, wrong secret) ×1; grover (right family, wrong secret) ×1.

Stress confusion matrix (rows = true family, columns = answer; `none` = no answer; 100 oracles per row):

true \ answer	const	bal	linear	grover	rand	simon	simon2	perm	none
constant	100	.	.	.	.	.	.	.	.
balanced	.	98	.	.	2	.	.	.	.
linear	.	.	100	.	.	.	.	.	.
grover	8	.	.	92	.	.	.	.	.
randfun	.	7	.	2	91	.	.	.	.
simon	.	.	.	.	.	94	.	6	.
simon2	.	.	.	.	.	2	98	.	.
perm	.	.	.	.	.	.	.	100	.

Reading: **grover → constant** is the dominant structural error: a 1-in-256 item that the 2-4 affordable Grover shots miss is indistinguishable from constant 0, and the posterior is right to fall back on it. **balanced ↔ randfun** is the weakest-signal pair: it rests on a zero-string count and one-bit statistics, with randfun's ones-fraction overlapping balanced only at its tails. At stress noise **simon → perm** occurs when readout noise makes violating samples common (see §2, perm bullet), so the data look uniform. Fixes we did not make: spending more shots on Grover costs queries on every constant-0 task; a lower tau would convert some of these into early wrong answers.

6. Where quantum helps

- **grover**: found with a few k=6-8 Grover shots plus two confirming reads (see budget table), against about 128 expected classical queries (a classical reader finds t with probability ~60/256 ≈ 0.23 in 60 queries; measured baseline accuracy in §4).
- **linear**: one `bv` shot gives a; with b by classical reads that is ~6 queries, versus 9+ noisy reads classically; `bv` also separates constant/linear/balanced/randfun at once.
- **simon/simon2**: a probe returns a linear equation per query; classical birthday search needs ~13-23 distinct reads before one collision and fails under misfires.
- **classical is enough** for constant 1/0 vs balanced, and for linear once a is known: reads give b and confirm. At n=8 the quantum advantage is moderate, not exponential.

7. Reproduce

`.venv/bin/python evaluate.py --suite all --reps 100 --workers 8` (about 2.2 h on 8 cores); `evaluate.py --suite tau --reps 40 --out logs_tau`; `compare_tau.py logs_tau`. Max per-oracle time 24 s (limit 120 s); max 16 qubits (limit 20); only qiskit/numpy/stdlib.